

PTVA's
MULUND COLLEGE OF COMMERCE
SEMESTER END TERM 1 EXAMINATION
FYBBA SEMESTER I 2023 - 24

Subject – Every Day Maths
Date – 03/11/2023

Time – 1 hrs
Marks – 30

Q1. Attempt the following

- a) Find the amount accumulated if I deposit Rs. 3500 per month at 7.5% for 3 years (08)
b) Find present value of annuity of Rs. 4000 payable yearly for 5 years at 14%. Per annum. (07)

OR

- c) I want Rs. 3000000 at the end of 4th year. How much amount should I deposit every 6 months at 6.5%.pa. (08)
d) Find EMI payable on Rs.90000 at 12%.pa for 4 years. (07)

Q2 Attempt ANY 3 of the following (15)

a) If $A = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 \\ 1 & 5 \\ 1 & 1 \end{bmatrix}$ find AB and BA Check whether $AB = BA$ (08)

b) if $A = \begin{bmatrix} -1 & 1 & -1 \\ 3 & -3 & 3 \\ 5 & -5 & 5 \end{bmatrix}$ show that $A^2 = A$ (07)

OR

c) Find A^{-1} if $A = \begin{bmatrix} 1 & 3 & 1 \\ 3 & 2 & 4 \\ 2 & 1 & 1 \end{bmatrix}$ (08)

d) i) Find the value of $\begin{vmatrix} 91 & 94 & 96 \\ 92 & 94 & 97 \\ 93 & 96 & 98 \end{vmatrix}$ (04)

ii) Show that $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 6 & 9 \end{vmatrix} = 0$ without expanding. (03)